Topic 10: Ecosystems

This topic considers the interactions between the organisms and the environment within an ecosystem. It includes details of how biotic and abiotic factors are involved in the development of ecosystems over time. Human influences on ecosystems are also discussed along with the need for conservation.

Opportunities for developing mathematical skills within this topic include: estimating results; constructing and interpresting frequency tables and diagrams, bar charts and histograms; understanding principles of sampling, including use of a formula to calculate an index of diversity; understanding the terms mean, median and mode; selecting and using statistical tests, including the Chi squared test, Student's t-test of difference, Spearman's rank, correlation coefficient; standard deviation and range. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

Students should:	
10.1	**The nature of ecosystems**
i	Understand what is meant by the term ecosystem and that they range in size.
ii	Understand what is meant by trophic levels.
iii	Understand the advantages and disadvantages of pyramids of numbers, biomass (dry) and energy as useful representations of ecosystem structure and how biomass and energy are transferred within them.
iv	Know the ecological techniques used to assess abundance and distribution of organisms in a natural habitat, including types of quadrat, transects, ACFOR scales, percentage cover and individual counts.
v	Be able to select appropriate ecological techniques according to the ecosystem and organisms to be studied.
CORE PRACTICAL 15: Investigate the effect of different sampling methods on estimates of the size of a population taking into account the safe and ethical use of organisms.	
vi	Be able to use statistical tests to analyse data, including t-test and Spearman rank correlation coefficient.